

THINKNOTE

Using retrieved knowledge to
revise an initial answer

A comparison with In-Context RALM, REPLUG, & SELF-RAG

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PROBLEM AND MOTIVATION

The Problem: Standard LLMs struggle when they are exposed to new or unfamiliar outside information.

The “Passive recipient” flaw: Standard RAG treats the AI like a student passively copying notes from a textbook, absorbing text without integrating it.

Key Failures:

Hallucination: Model may give incorrect answers when answers aren't quoted verbatim.

Premature Defeat: Models "give up" if explicit text matches aren't found.

Search Distraction: Off topic information can confuse the model and weaken its reasoning.

MAIN IDEA

Standard retrieval often adds documents to the prompt and hopes the model will use them correctly.

THINKNOTE instead asks the model to build a temporary note from the retrieved knowledge, inspect its own reasoning, and revise its answer. It does this in two stages

1) Knowledge Assimilation:

$$\mathcal{T} = \mathcal{M}(\mathcal{I}_{\text{ka}}, q, \mathcal{D}),$$

1.5) Self Inquiry:

$$\mathcal{R} = \mathcal{M}(\mathcal{I}_{\text{coT}}, q).$$

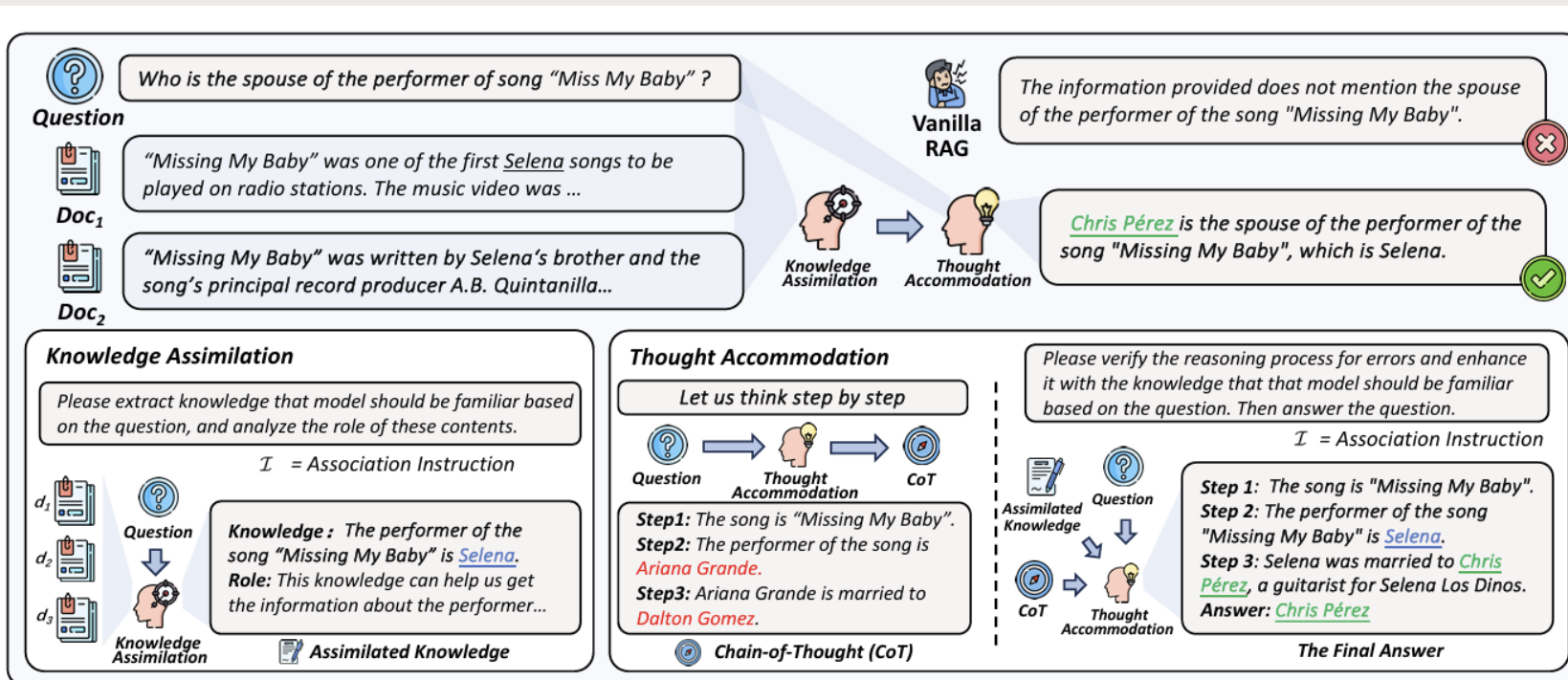
2) Thought Accommodation:

$$y = \mathcal{M}(\mathcal{I}_{\text{ta}}, q, \mathcal{T}, \mathcal{R}),$$

EXAMPLE

- The note identifies the performer and connects this fact to what the question is really asking.

- Without integration, a model may focus on a wrong performer and answer confidently but incorrectly.



- Thought accommodation corrects the reasoning path before giving the final response.

EXPERIMENTS AND RESULTS

- THINKNOTE achieved the strongest overall performance. Evaluation used 500 randomly sampled questions from each benchmark dataset due to inference cost.
- It consistently outperformed Direct QA, Chain of Thought, Vanilla RAG, and self refined baselines.
- With Llama-3-Ins 70B, THINKNOTE reaches 69.8% on PopQA, 68.4% on NQ, and 85.4% on TriviaQA.
- With ChatGPT-4o mini, THINKNOTE also performed strongly, including 51.5% on ASQA string exact match and 61.3% on ASQA recall.
- The results suggest that actively assimilating retrieved knowledge and revising reasoning is more effective than simply feeding the model retrieved documents.

Method	LLM	PopQA (acc)	NQ (acc)	TriviaQA (acc)	2WikiMHQA (acc)	(rec)	ASQA (str-em)	(hit)	(rec)
Vanilla LLMs									
Direct QA	Llama-3-Ins _{8B}	24.2	39.2	67.2	43.0	48.1	24.8	5.0	16.1
	Llama-3-Ins _{70B}	34.2	54.2	80.4	53.8	60.0	33.3	8.8	20.2
	ChatGPT-4o _{MINI}	32.6	51.0	75.0	47.4	52.3	31.4	7.8	18.5
CoT	Llama-3-Ins _{8B}	24.8	44.0	69.4	47.2	51.4	28.8	7.8	25.5
	Llama-3-Ins _{70B}	31.6	54.4	80.6	56.4	62.6	36.4	11.2	32.7
	ChatGPT-4o _{MINI}	32.4	53.2	76.2	51.0	55.3	32.4	8.0	21.6
LLM w/ RAG									
Vanilla RAG	Llama-3-Ins _{8B}	59.8	56.0	71.2	45.2	49.3	34.2	12.0	27.7
	Llama-3-Ins _{70B}	63.4	59.8	75.2	52.2	56.3	38.1	14.6	29.8
	ChatGPT-4o _{MINI}	62.0	64.8	77.8	51.4	56.2	38.9	13.2	28.0
Self-Refined	Self-RAG _{7B}	53.8	43.4	66.4	36.6	40.2	28.6	9.0	13.2
	Self-RAG _{8B}	54.8	46.2	67.0	37.2	42.6	37.6	15.6	21.4
	Chain-of-Note _{8B}	64.2	59.8	73.0	52.0	56.2	42.0	17.8	56.2
	Chain-of-Note _{70B}	68.2	67.8	78.4	56.8	61.1	45.0	20.6	54.9
THINKNOTE	Llama-3-Ins _{8B}	65.8	62.0	79.8	53.8	60.0	43.4	18.8	57.8
	Llama-3-Ins _{70B}	69.8	68.4	85.4	63.2	68.4	48.8	22.6	59.9
	ChatGPT-4o _{MINI}	69.8	71.0	83.4	61.0	67.4	51.5	24.6	61.3

Table 1: Overall Performance of THINKNOTE and Baselines on Different Knowledge-Intensive Benchmarks. For THINKNOTE, we use the instruction of association learning behavior to guide both knowledge assimilation and thought accommodation and evaluate the overall performance.

EXPERIMENT 2

- Both THINKNOTE stages are important.
- Removing Knowledge Assimilation causes the biggest performance drop.
- Thought Accommodation also improves final answer quality.
- Association performs the strongest compared to other instruction strategies.
- Results suggest the full two stage framework works better than either component alone.

Method	Llama-3-Ins-8B			Llama-3-Ins-70B		
	PopQA (acc)	2WikiMHQA (acc)	ASQA (str-em)	PopQA (acc)	2WikiMHQA (acc)	ASQA (str-em)
Cognitive Modeling Ablation						
THINKNOTE	65.8	53.8	43.4	69.8	63.2	48.8
w/o Knowledge Assimilation	58.4	48.2	38.6	61.8	59.6	43.2
w/o Thought Accommodation	63.0	46.8	40.2	66.0	57.4	45.7
Learning Behaviors Ablation						
w. Anchoring Instruction	64.8	54.6	39.9	69.0	63.6	45.8
w. Association Instruction	65.8	53.8	43.4	69.8	63.2	48.8
w. Reasoning Instruction	64.0	52.8	40.3	68.4	59.6	48.3
w. Reflection Instruction	63.2	52.8	42.2	65.2	60.6	50.2

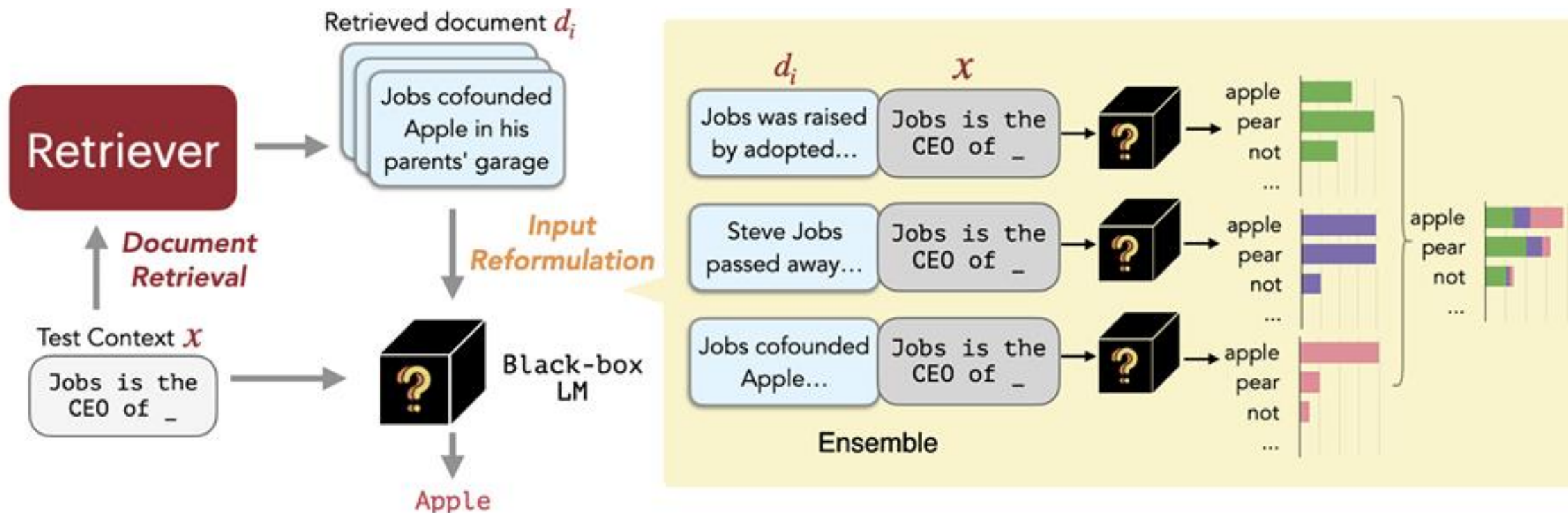
Table 2: Ablation Study of THINKNOTE. We examine the contributions of knowledge assimilation and thought accommodation, and vary instruction designs to simulate different learning behaviors. Experiments are conducted with Meta-Llama-3-Instruct-8B and Meta-Llama-3-Instruct-70B.



Figure 1: An example of *In-Context RALM*: We simply prepend the retrieved document before the input prefix.

IN-CONTEXT RALM

- Retrieves a relevant external document.
- Prepends the document to the LM input.
- Uses an existing, frozen language model.
- No additional LM training is required.

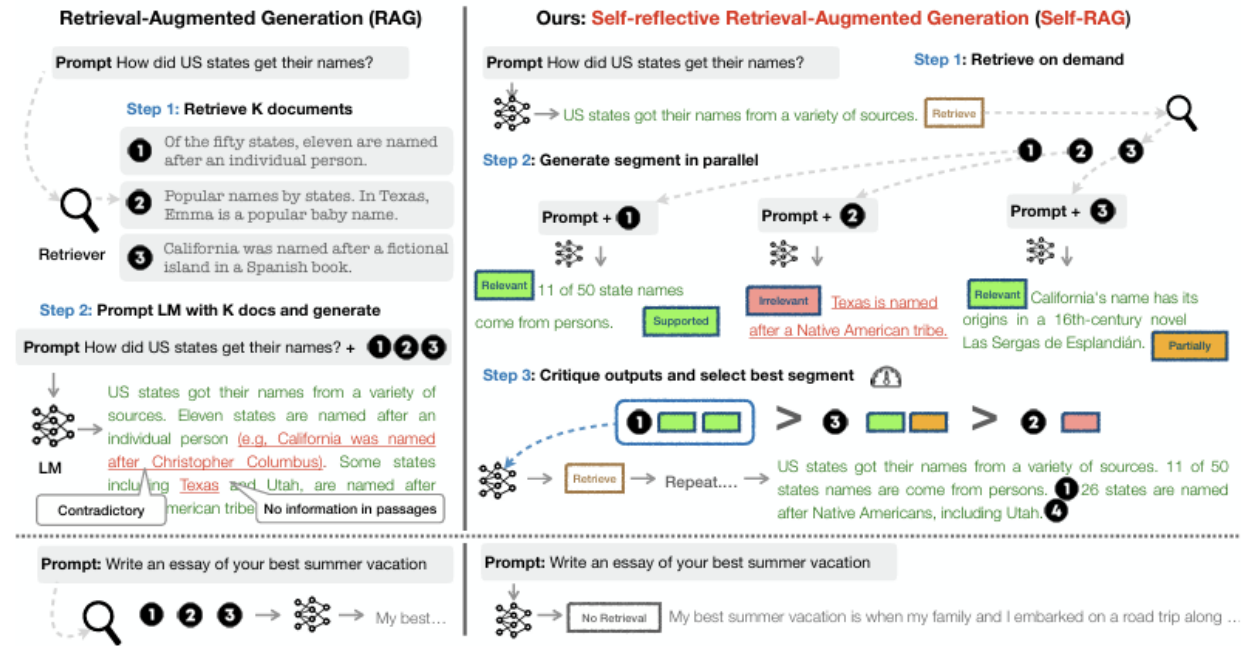


REPLUG

- Retrieves multiple relevant documents.
- Processes each document separately with the same frozen LM.
- Combines the outputs from separate model runs, each using a different retrieved document, to make the final prediction.
- Works with a black box LM without modifying its parameters.

SELF – RAG: RETRIEVE, GENERATE, CRITIQUE

- Retrieves on demand instead of always retrieving a fixed number of documents.
- Evaluates retrieved passages for relevance and whether they support the generated answer.
- Uses reflection tokens to critique outputs and select the best response.



COMPARATIVE STRENGTHS

- No LM retraining required: Unlike SELF-RAG, THINKNOTE can be applied through prompting without training the model to use special reflection tokens.
- Constructivist approach: It mirrors how students revise understanding after reading new material.
- Strong experimental results: According to the experiments conducted by the authors, THINKNOTE achieved a 10% improvement in comparison to other baseline methods on various question- answering benchmarks
- Adaptability : THINKNOTE can work across different types of LLM's and retrieval setups.
- More active use of retrieved knowledge : Instead of only filtering or inserting retrieved text, THINKNOTE tries to assimilate it with the model's existing knowledge before producing the final answer.

COMPARATIVE WEAKNESSES

- Higher cost than simpler RAG methods: THINKNOTE needs the model to run multiple times before producing the final answer, while methods like In-Context RALM can simply add the retrieved document to the prompt.
- Less control over retrieval than SELF-RAG: SELF-RAG can decide when retrieval is actually needed, while THINKNOTE mainly focuses on what to do with the information after it has already been retrieved.
- Less focus on checking the evidence: SELF-RAG checks whether the retrieved information is relevant and whether it supports the final answer, while THINKNOTE focuses more on assimilating the information into its reasoning.
- More limited testing: THINKNOTE was mainly tested on tasks which involved answering questions, while SELF-RAG is tested across a broader range of tasks.

POSSIBLE EXTENSIONS

Multi document summarization: Use knowledge assimilation to combine important facts from several documents, then use thought accommodation to revise an initial summary for accuracy and completeness.

Fact checking / claim verification: Retrieve evidence about a claim, connect that evidence with the model's existing knowledge, and revise an initial judgment when the evidence contradicts it.

Planning and agent systems: An AI agent could assimilate new information from searches or tools and then revise its original plan instead of continuing with outdated assumptions.

Combine THINKNOTE with SELF-RAG: SELF-RAG could decide when retrieval is needed, while THINKNOTE could determine how the retrieved information should be integrated into the reasoning.

CONCLUSION AND QUESTIONS

THINKNOTE's main contribution is a more active way of using retrieved information: the model assimilates evidence, generates its initial reasoning, and then revises its answer.

This paper is important because it shifts the focus from “Did we retrieve documents?” to “Did the model actually use them?”

Questions?

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